

AY 2025/2026



POLITECNICO
MILANO 1863

Internet Of Things: Challenge 3

Part 2

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Version 1.0
May 11, 2026

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1 Exercise

1.1 EQ1

A LoRaWAN network in Europe operates at carrier frequency 868 MHz with bandwidth 125 kHz. The network is composed of one gateway and 40 sensor nodes. Each sensor node transmits packets with payload size 20 bytes according to a Poisson process with intensity $\lambda = 2$ packets/minute.

Determine the largest LoRa spreading factor (SF) that guarantees a packet success rate of at least 75%. Assume that all nodes use the same SF and adopt a simplified model where packet transmissions are uncoordinated and may overlap in time. To compute the airtime of a packet, use the TTN airtime calculator: <https://www.thethingsnetwork.org/airtime-calculator>

Energy Costs

$$\begin{aligned} F &= 868 \text{ MHz} \\ BW &= 125 \text{ kHz} \\ n &= 40 \text{ nodes} \\ \lambda &= 2 \text{ packets/minute} \end{aligned}$$

The total amount of messages is

$$\lambda_{tot} = n \cdot \lambda = 40 \cdot 2 = 80 \text{ packets/minute}$$

Now the success probability is

$$P_{success} = e^{-2 \cdot \lambda_{tot} \cdot T_a}$$

We want that to be $\geq$ than 0.75

$$P_{success} = e^{-2 \cdot \lambda_{tot} \cdot T_a} \geq 0.75$$

Solving for T_a we get

$$T_a < 0.107 \text{ s} = 107 \text{ ms}$$

Now, from the airtime calculator, we get this airtime.

The screenshot shows the TTN Airtime Calculator interface. At the top, there are four input fields: 'Input Bytes' (20), 'Spreading Factor' (SF7), 'Region' (EU868), and 'Bandwidth' (125 kHz). Below these fields, a large blue box displays the result '71.9 ms'. Underneath the blue box, the text 'Time on air' is visible.

Figure 1: SF7

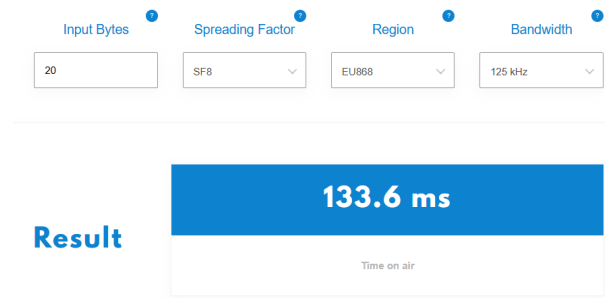


Figure 2: SF8

So the largest LoRa spreading factor that guarantees a packet success rate of at least 75% is SF7.

1.2 EQ2

After deploying the network in the field, the measured packet success rate is significantly lower than expected. Moreover, performance is not uniform across nodes: some nodes achieve relatively good delivery rates, while others experience very poor performance.

I would move the nodes closer to the gateway so that we would improve the signal-to-noise ratio for the nodes with very poor performance.